EX NO: 9 SIMULATION OF CONVOLUTIONAL CODES AND ANALYSIS OF ENCODED OUTPUT BIT SEQUENCES USING MATLAB

Objectives:

(i) To implement convolutional encoding for a given binary input sequence using a rate convolutional encoder and tabulate the corresponding encoded output pairs.

(ii) To visualize and compare the input bit sequence and encoded output bit sequence using MATLAB plots to analyze the redundancy introduced by convolutional coding.

MATLAB code:

```
clc;
clear;
close all;
% Fixed input sequence
input_bits = [1 0 0 1 1];
% Convolutional encoder
trellis = poly2trellis(3,[7 5]);
encoded_bits = convenc(input_bits,trellis);
% Separate encoded bits into pairs
encoded_pairs = reshape(encoded_bits,2,[])';
% Create output strings
for i = 1:length(input_bits)
output_pair{i} = sprintf('%d%d', ...
encoded_pairs(i,1),encoded_pairs(i,2));
end
% Create table
Bit_Position = (1:length(input_bits))';
Input_Bit = input_bits';
Output_Encoded = output_pair';
```

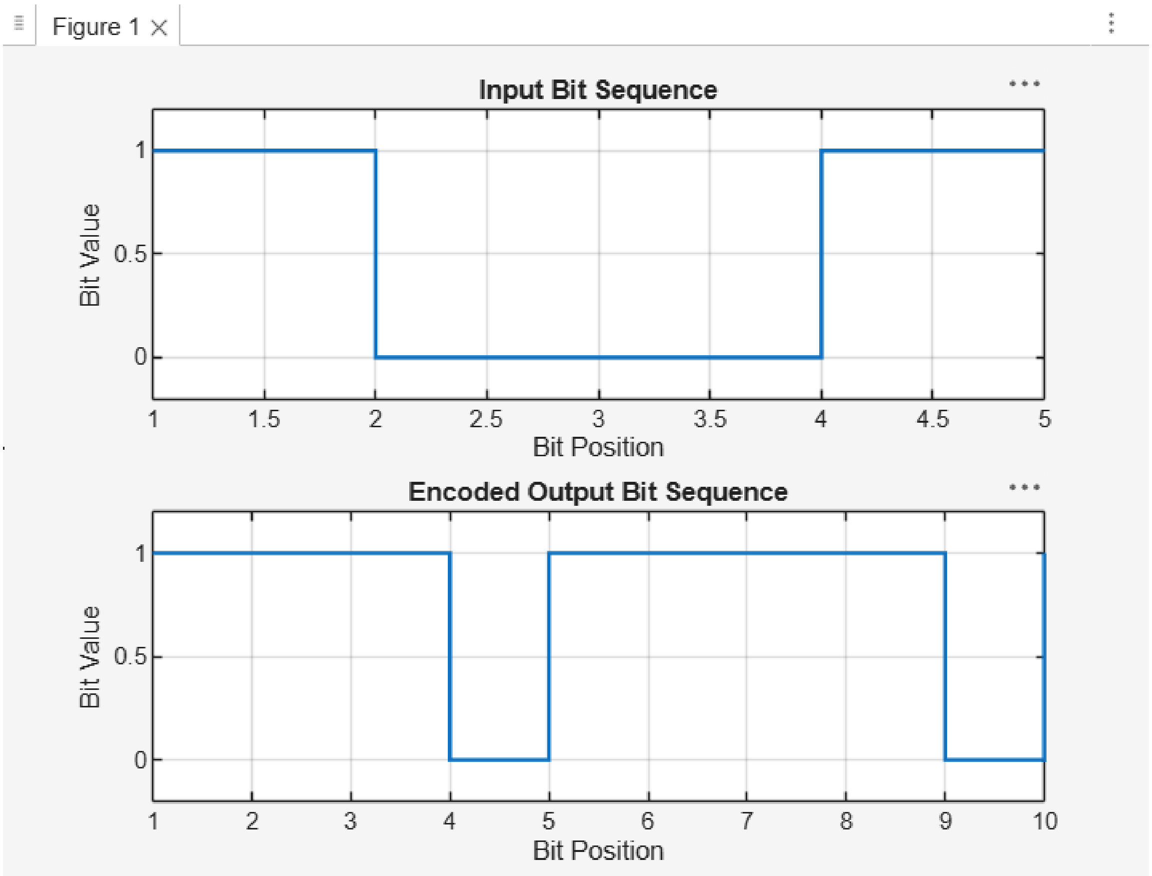
```
T = table(Bit_Position,Input_Bit,Output_Encoded);
disp('Convolutional Encoding Table');
disp(T);
% Plot
figure;
subplot(2,1,1)
stairs(input_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Input Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
subplot(2,1,2)
stairs(encoded_bits,'LineWidth',1.5)
ylim([-0.2 1.2])
title('Encoded Output Bit Sequence')
xlabel('Bit Position')
ylabel('Bit Value')
grid on
```

OUTPUT:

Convolutional Encoding Table

Bit_Position	Input_Bit	Output_Encoded
1	1	{'11'}
2	0	{'10'}
3	0	{'11'}
4	1	{'11'}
5	1	{'01'}

>>



RESULT:

The convolutional encoder was successfully implemented using a constraint length of 3 and generator polynomials . The input bit sequence was encoded into the output sequence 1110111101, and the corresponding encoded output pairs were tabulated successfully**.**